1 **B**

$$\rho = \frac{m}{V} = \frac{m}{\frac{4}{3}\pi r^3} = \frac{80}{\frac{4}{3}\pi (2.0)^3} = 2.3873 \text{ g cm}^{-3}$$

$$\frac{\Delta\rho}{\rho} = \frac{\Delta m}{m} + 3\frac{\Delta d}{d} \quad \left(\text{or } \frac{\Delta\rho}{\rho} = \frac{\Delta m}{m} + 3\frac{\Delta r}{r}\right)$$

$$\frac{\Delta\rho}{\rho} = \frac{2}{80} + 3\frac{0.1}{4.0}$$

$$\Delta\rho = 0.24 = 0.2 \text{ (1 s.f.)}$$

$$\rho = (2.4 \pm 0.4) \text{ g cm}^{-3}$$

2 **C** Between t_1 and t_2, displacement increases at increasing rate. Between t_2 and t_3